

Collaboration Profiles and their Impact on Musical Success

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ABSTRACT

We identify collaboration profiles in a musical network composed of successful artists. We argue that the way in which artists professionally connect with each other can significantly impact their success. Using data from Billboard and Spotify, we construct a collaborative success-based network to identify different profiles, as well as to analyze their impact on artists' popularity. Through topological metrics and clustering algorithms, we identify three well-defined communities with distinct collaboration patterns and notable discrepancies in levels of musical success. We have found that successful artists are more likely to have profiles with a high degree of interaction and high diversification. These findings offer a new perspective on success in the music industry, unraveling how collaboration profiles can contribute to an artist's popularity.

CCS CONCEPTS

• Information systems → Social networks;

KEYWORDS

Collaboration profiles; creative collaboration; music success; social networks

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1 INTRODUCTION

Although social interaction is frequently thought to increase creativity and drive success in the music industry, there is still controversy about the optimal structure and relative benefits of collaboration. For example, DJ Khaled (an American hip-hop music producer) is often criticized for the merits of his collaborative music with acclaimed artists. However, no one can deny that he is today one of the most successful hip-hop artists in the world. In just over a decade, Khaled has accumulated 24 hits on the Billboard Hot 100 (a

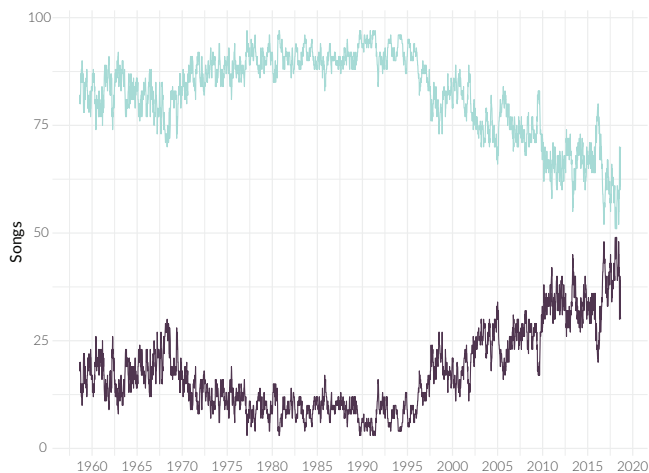


Figure 1: Billboard Hot 100 songs (1958 - 2018). The lighter line represents hits without collaborations, while the darker reflects musical collaborations.

ranking with the week's most popular songs across all genres), all of them collaborating with other artists¹.

Khaled's method is not innovative. In fact, according to data from the Billboard Hot 100, the number of songs executed by more than one artist has been experiencing a significant increase in recent years, as summarized in Figure 1. In 2018 alone, these collaborations already represent more than 40% of the hit songs. Although they can yield the acclaim of fans and the top rankings when well executed, such connections often run the risk of sounding artificial and inauthentic as defined by professional critics (e.g., the aforementioned DJ Khaled's case).

Even with the growing popularity of musical collaborations, factors that lead to the success of a collaborative process are not entirely understood [16]. Therefore, investigating how music collaboration profiles can positively impact an artist's popularity is still worth pursuing. For instance, while talent and status attract social connections, the researchers ignore that social networks can independently promote success [12]. Previous research in music collaboration networks has established that social interaction can influence the trajectory of the music industry [4, 7, 19]. On a broader scale, Budner and Grahl [4] found that different collaborative roles hold a distinct impact on network connectivity. Moreover, the influence of a specific function depends not only on the number of relationships but also on the importance of these connections.

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¹"Popular music is more collaborative than ever", *The Economist*, (August 25, 2018), <https://www.economist.com/graphic-detail/2018/02/02/popular-music-is-more-collaborative-than-ever>

To further our understanding of how music collaboration profiles affect artists' popularity, we analyze and identify such characterizations in a musical success-based network; that is, a network composed only of successful artists. By detecting communities and their respective patterns of network collaboration, we focus on analyzing the impact of these profiles on successful musical artists. Considering topological metrics, we define four main categories of collaboration profiles: *Interaction*, *Distance*, *Influence* and *Similarity*. Among them, we find that the first three affect musical success more intensely than *Similarity*. These findings provide evidence that (i) there are indeed distinct success factors for music collaboration profiles that are socially measurable, and (ii) there are common factors to successful collaboration in the music market.

The remainder of this paper is organized as follows. In Section 2, we discuss related work. In Section 3 are fundamental concepts used in this work. In Section 4, we describe the dataset and introduce the proposed methodology. We detail the results and experimental evaluation in Section 5. We conclude with future research directions in Section 6.

2 RELATED WORK

Computer Science studies over the music industry typically fall within the areas of recommendation systems [20], human-centered computing [9] and music retrieval [11]. Nevertheless, ongoing efforts still try to solve an interesting, complex problem: *predicting musical success*. Over the years, the number of artists and musical productions has considerably increased; and so have the number of attempts to discover the recipe for turning a song into a hit. Indeed, there are plenty of analyses on factors that potentially influence musical success from varied perspectives. Examples include: using metrics of social media [3], investigating the relationship between user-generated Internet content and album sales [1, 6], and analyzing the musical characteristics [2].

Specially based on social media metrics, Bischoff et al. [3] introduced an approach to predict the potential of songs to become hits. They show the possibility for accurately predicting the commercial success of a song (81.31% success rate) from the relations generated in the topological structure of a social network. Their findings show that observing network metrics may provide some indication of music sales. However, in another context, Dewan and Ramaprasad [6] later drew a parallel between new media, traditional media and music industry sales. Their findings indicate a direct correlation between radio plays and commercial success, but there is no such correlation between social media activity and increased album sales. In a more recent study, Araujo et al. [1] performed a sentiment analysis on Twitter comments for 30 days prior to the album release. They also correlated positive comments with musical success using popularity measures available on Spotify and Billboard. Once again, no evidence was found that the volume of comments on Twitter is related to the album's success.

On a broader perspective, explaining or predicting the success of creative individuals through social network analysis has been a hot topic for decades. In 1973, Granovetter [8] suggested that the topology of an individual's social network has an impact on the personal success. Following approaches have focused on social collaborations [4, 5, 19] as well. In the context of our work, the

connection between network topology and success was also found in the musical context. Uzzi and Spiro [19] analyzed a network of creative collaboration among Broadway musical artists from 1945 to 1989. By applying statistical methods, they found that network measures significantly affected creativity in terms of the financial and artistic success. In another setting, Budner and Grahl [4] investigated the large-scale structure of musical collaborations by employing tools of network science. The authors seek to fill in the lack of knowledge by examining the impact of various groups of collaborators on the size of the giant component of a musical collaboration network. In a more recent study, Calefato et al. [5] investigated creative collaboration in a music community where authors compose songs together through overdubbing. The authors analyzed the relationship between metrics related to song- and author-related measures and the likelihood of a song being overdubbed.

Nevertheless, such papers ignore the hypothesis that the popularity of artists may be directly related to their *collaborative patterns*. There is still no systematic study on the influence of interaction profiles on musical success. In other words, there is still space for deeply examining the numerous factors of collaboration and creative success. In fact, most existing proposals are limited to analyzing non-topological metrics referring to the music or the artist, as well as topological measures related only to the nodes' connections. Therefore, there is a strong potential for modeling success through the social interaction profile in the music industry.

3 FUNDAMENTAL CONCEPTS

In our study, we consider well known metrics dependent on the node (Clustering Coefficient, Eigenvector, Degree and Weighted Degree) and related to the whole graph (Closeness, Eccentricity, and Betweenness), summarized as follows.

Clustering Coefficient. Also known as the transitivity measure, it captures the degree to which the neighbors of a given node bind to each other. In other words, it measures the density of local links in the graph: the denser interconnected the neighborhood of a node i , the greater the local clustering coefficient C_i .

Eigenvector Centrality. It is a measure of the influence of a node within a graph. It calculates the centrality degrees of each node in the graph, but not always with the same equivalence, as the central nodes that connect to other central nodes receive more weight than those that do not join to central nodes.

Degree and Weighted Degree. Degree is the number of edges incident on a given node. Weighted degree is defined similarly by summing the weights of the incident edges. In our network, the degree measures the number of connections to other artists (nodes), while the weighted degree measures the number of interactions.

Closeness Centrality. It is a node centrality index based on the shortest path. Specifically, the value given to each node corresponds to the average of the shortest path between that node and all others in the graph. Notably, high (low) values of closeness should indicate that all other nodes are close (distant) to the one being measured.

Eccentricity. It is also a node centrality index. For each node n , it calculates the shortest path between n and all other nodes in

the graph, then the “longest” shortest path is chosen. Having such path identified, its reciprocal is calculated. High values indicate a positive meaning in terms of node proximity: if the eccentricity of n is high, all other nodes are close; if it is low, there is at least one node (and all its neighbors) that is far from n .

Betweenness Centrality. Another way to measure the node centrality using the shortest path by considering how many times a given node appears in the shortest path of the other nodes in the graph. Thus, a high betweenness score means that the node is crucial to maintaining node connections for certain paths. Such connectors have the potential to be highly influential by inserting themselves into the dealings of other parties.

4 METHODOLOGY

This section presents our methodology for analyzing the impact of collaboration profiles on musical artists’ success. First, we describe the process of building a dataset in Section 4.1. From such data, we model a social network in Section 4.2. Then, we introduce a method to detect the collaboration profiles in Section 4.3, which is based on the topological metrics summarized in Section 3. Finally, we explain our approach to investigate if collaboration patterns can influence musical success in Section 4.4.

4.1 Data Collection

Billboard is a weekly American magazine specialized in music. Its website provides countless internationally recognized rankings that classify songs and popular albums. To model a success-based network of collaborations, we collected all the artists on Billboard’s Artist 100² chart, a weekly ranking that lists the top 100 artists. We use the *billboard.py*³ Python API for access Billboard’s rankings and perform the data collection. In total, we collected 211 rankings between 2014 (July 26, 2014) and 2018 (July 28, 2018). As each chart consists of 100 artists’ names, we collected 21,100 artists’ names in total, which were deduplicated to achieve 1,135 distinct names.

For more information on each artist, we also collected data from Spotify, one of the most popular and used music streaming platforms. With Spotipy library⁴, we obtained full access to all of the music data provided by the Spotify platform. Then, for each artist collected on the Billboard chart, we also collected features and her/his ten most popular songs. In the same way, the artists (and their features) that collaborate in the execution of the top 10 songs were also collected. Therefore, the total number of artists grew to 2,152. For each artist, the following information was recovered⁵: (i) the Spotify ID, (ii) name, (iii) popularity (a numeric value between 0 and 100, with 100 being the most popular), (iv) the total number of followers and (v) a list of the genres the artist is associated with.

4.2 Social Network Modeling

To model the music collaboration network, we collected all the artists who participated in the execution of a single —either as participation (featuring) or collaboration (with). This is because, aside from Spotify not distinguishing the artists’ role in a song execution,

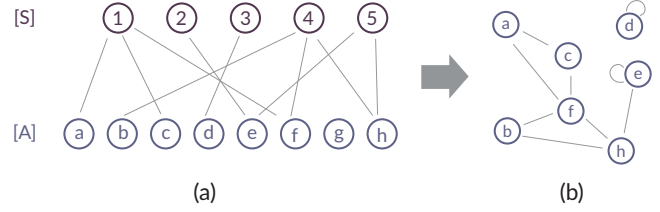


Figure 2: Bipartite graph projection: the two-mode social network for songs S_{1-5} and artists A_{a-h} (a) is projected onto a single one-mode social network (b).

Table 1: Collaboration social networks’ statistics

	Original Network	Filtered Network
# Artists	2, 152	354
# Songs	10, 706	2, 144
# Collabs	5, 335	922
Modularity	0.793	0.478

all the artists who collaborated in it represents an essential part in the success of a single. The dataset was modeled as a bipartite graph with nodes for a set of songs and a set of artists, and edges connecting the individuals who collaborated in the execution of each song. As most network analysis techniques cannot be applied to bipartite graphs, we project the social network into a one-mode model, as explained next.

In the bipartite graph, there are two groups of nodes: S the set of songs; and A the set of successful artists who collaborated in the execution of the songs present in S . Following the methodology proposed in [13], the bipartite model was then designed as a uni-modal non-directed graph: every two nodes in A are connected by a link if they are connected to the corresponding node in S (in the bipartite representation). In other words, only artists are present as nodes, and edges exist between artists who worked on the same song in such a projection.

In addition, songs that do not have more than one artist in their execution are modeled as self-loops on the artist’s node. Figure 2 shows this projection. Before proceeding to the next step, we perform a network filtering for keeping only successful artists (See definition in Section 4.4) on the network and getting rid of potential outliers. Therefore, the network includes only artists with a popularity rating greater or equal than 70, and at least 1,000,000 followers. Table 1 presents statistics of the originally collected network and the new, filtered one: (i) the total number of artists, (ii) total number of songs, (iii) total number of musical collaborations, and (iv) network modularity.

4.3 Collaboration Profiles

From the topological metrics (Section 3), we define four categories of collaboration profiles, each with two levels of subcategories. The classes are defined according to the specific social metrics’ features as follows.

- (1) **Interaction** (degree and weighted degree)
 - (a) Single artists collaboration
 - (b) Multi-artists collaboration

²Billboard’s Artist 100: <https://www.billboard.com/charts/artist-100>

³*billboard.py*: <https://github.com/guoguo12/billboard-charts>

⁴*Spotipy*: <http://spotipy.readthedocs.io/en/latest/>

⁵Full dataset openly available at http://bit.ly/apoena_datasets

- (2) **Distance** (closeness and eccentricity)
 - (a) Nearby collaboration
 - (b) Distant collaboration
- (3) **Similarity** (clustering coefficient)
 - (a) Inter-genre collaboration
 - (b) Intra-genre collaboration
- (4) **Influence** (betweenness and eigenvector)
 - (a) Non influential collaboration
 - (b) Influential collaboration

The first category, called *Interaction*, is based on the artist (node) connectivity and is directly related to Degree and Weighted Degree metrics (quantify the number of connections/interactions of a node). High values of these metrics suggest a collaboration composed by more than one artist; likewise, low values indicate non-collaborative artists. Thus, the category ranges from single-artist (or band) to multi-artist collaborations.

The *Distance* category is based on proximity by Closeness and Eccentricity metrics. Note that closeness should always be compared to Eccentricity [15]: a node with high eccentricity and high closeness is very likely to be central in the graph. In fact, the values of such metrics can be considered as an “average tendency to node proximity or isolation” [15]. Therefore, high values indicate pivotal artists in the graph, which are close to the other nodes. We also consider this proximity to physical distance (the same country, city, state, etc.); i.e., the artist can have a nearby (low degree of proximity, therefore less central in the network) or an distant (high degree of proximity, more central) collaboration profile.

Similarity is the most complex category and related to the Clustering Coefficient (captures the density of edges in the neighborhood of a node/artist). The more local links within the neighborhood of a node, the higher its local clustering will be. In our case, a link represents a musical collaboration. Therefore, an edge can be defined by a collaboration between artists who are present in the same professional/social circle. It is safe to say that these associations are usually made up of artists from the same musical genre, where they often share the same producer, events, audience, and so on. Consequently, a high clustering coefficient indicates similar connections (probably, of the same musical genre); furthermore, low values imply in more diverse collaborations and, in general, between different genres.

Finally, the *Influence* category, as the name already says, incorporates the concept of network influence. Betweenness and Eigenvector metrics accurately explore this concept, quantifying the importance of a node in a social network. A node with high values of these metrics has the potential to be highly influential, having access to different network regions not reachable by other vertices. For that reason, the category is composed of influential and non-influential artists.

4.4 Statistical Analysis

Finally, we perform a statistical correlation analysis to evaluate the association between collaboration profiles and the artist’s success. To do so, we first must define success, which could be done through popularity on social media and music platforms, sales profit, awards, etc. Here, we define it as: *a successful artist presents both a high level of popularity and a large number of followers*. Formally, we define

that the success s_i of an artist i is high if his/her popularity index $p_i \geq 70$ and number of followers $f_i \geq 1,000,000$.

Now, to check if there is a correlation between the success of an artist and the network metrics, we consider a set of analyses from different perspectives. First, we perform the normality test of Shapiro-Wilk [17] over the the whole dataset to verify if they follow a normal distribution. Then, we apply three correlation measures. Note that the correlation coefficient is a widely used metric for the association between two sets of data. There are many different algorithms for calculating correlation, and each has properties and variants. Here, we use the three very common metrics, as follows. Finally, to determine the statistical significance level of each test, we also compute the t-test.

Pearson’s correlation (r) [14] measures a linear dependence between two variables (x and y). The coefficient (r) varies between -1 (perfect negative correlation) and 1 (perfect positive correlation). The value 0 indicates there is no linear correlation, but does not guarantee the independence between the variables.

Spearman’s correlation (ρ) [18] does not give any assumptions about the distribution of the data, but analyzes the appropriate correlation when the variables are measured on a scale that is at least ordinal. The method calculates the correlation between the classification of variables x and y . Its interpretation is similar to Pearson’s: the closer ρ is to ± 1 , the stronger the monotonous relationship is.

Kendall’s correlation (τ) [10] is a nonparametric test that measures the statistical dependence between two variables (x and y). Intuitively, the τ correlation between two variables will be high when the observations return a similar (or identical, with $\tau = 1$) classification between the two variables, and low when the observations have a different (or completely different, with $\tau = -1$) classification between the two variables.

5 RESULTS

Our goal is to identify collaboration profiles and assess their impact on the success of musical artists. To do so, we need to detect groups of artists and their respective collaborative patterns. Hence, we calculate the network metrics described in Section 3, and apply a clustering algorithm in Section 5.1. Then, we perform the statistical correlation analysis to evaluate the correlation between collaboration profiles and the artist’s success measure in Section 5.2.

5.1 Identifying Collaboration Profiles

With the collaboration profiles defined in Section 4.3, we need to identify which profiles actually exist in our dataset and the distribution of artists among them. Therefore, after calculating the topological metrics of each artist, we applied a clustering algorithm to group those with similar topological features. In this way, we find groups of artists with similar topological values and, consequently, similar collaboration profiles.

Note that there are different clustering algorithms that can be applied to finding subgroups of observations within a data set. Here, we use K-means, which is the simplest and most commonly used cluster method for dividing a dataset into a set of k groups. One of the first steps when using this algorithm is to define the number of

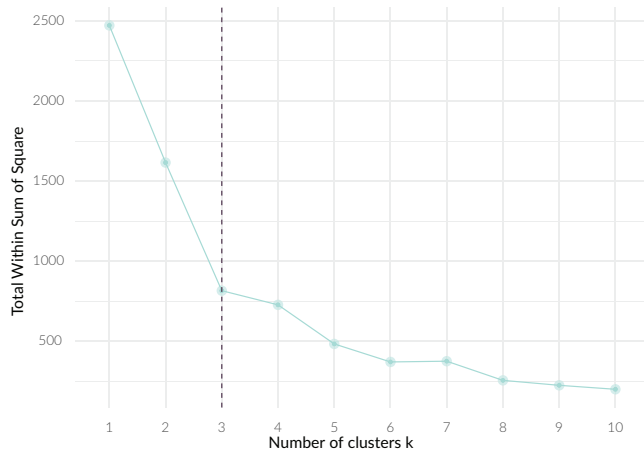
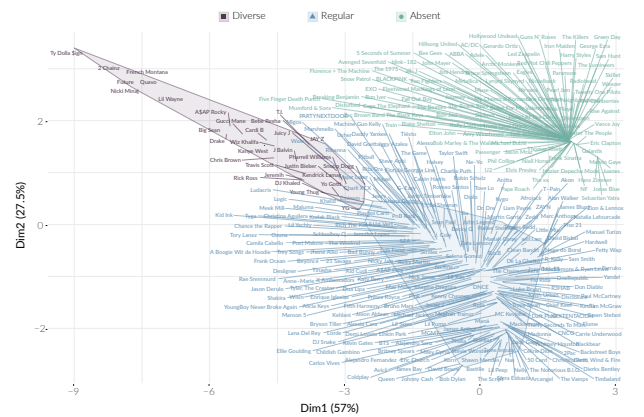
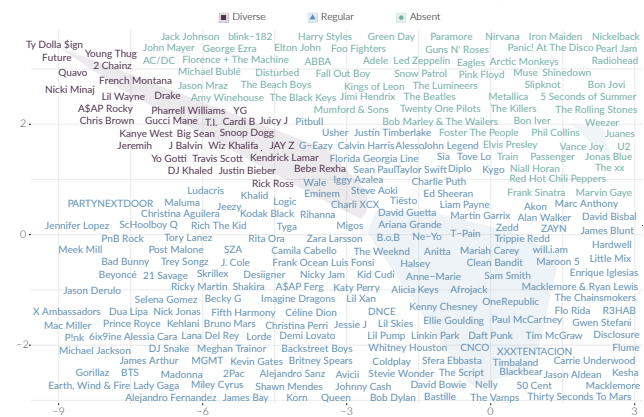


Figure 3: The optimal number of clusters ($k = 3$), according to Elbow Method.



(a) The result of complete clustering



(b) Example of some artists in each cluster

Figure 4: The partitioning k-means Clustering Plot.

Table 2: Collaboration Profiles

Profiles	Interaction		Distance		Genre		Influence	
	Degree	Weighted Degree	Eccentricity	Closeness	Clustering	Betweenness	Eigencentrality	
1A 2A 3A 4A	0	0	0	0	0	0	0	
1A 2A 3A 4B	0	0	0	0	0	1	1	
1A 2A 3B 4A	0	0	0	0	1	0	0	
1A 2A 3B 4B	0	0	0	0	1	1	1	
1A 2B 3A 4A	0	0	1	1	0	0	0	
1A 2B 3A 4B	0	0	1	1	0	1	1	
1A 2B 3B 4A	0	0	1	1	1	0	0	
1A 2B 3B 4B	0	0	1	1	1	1	1	
1B 2A 3A 4A	1	1	0	0	0	0	0	
1B 2A 3A 4B	1	1	0	0	0	1	1	
1B 2A 3B 4A	1	1	0	0	1	0	0	
1B 2A 3B 4B	1	1	0	0	1	1	1	
1B 2B 3A 4A	1	1	1	1	0	0	0	
1B 2B 3A 4B	1	1	1	1	0	1	1	
1B 2B 3B 4A	1	1	1	1	1	0	0	
1B 2B 3B 4B	1	1	1	1	1	1	1	

clusters to work with. We use a common solution to identify the optimum number of clusters: the Elbow method. It runs k-means on the dataset for a range of values of k (e.g., k from 1 to 10); for each value of k , it calculates the sum of squared errors (SSE); it then considers a line chart of the SSE for each value of k ; finally, if the line looks like an arm, then the “elbow” on the arm is the best k value. Figure 3 has the results for our dataset, and there is a clear bend (or “elbow”) at $k = 3$. This bend indicates that additional clusters beyond the third one would negatively affect the results by increasing k . Therefore, $k = 3$ is our optimal number of clusters.

Figure 4 shows the clustering results by using different colors according to each cluster assignment. According to Figure 4a, there is a natural division of three clusters: *Cluster 1* in the lower right corner; *Cluster 2* in the opposite upper corner to the left; and *Cluster 3* in the upper right corner. Note k-means clusters data into k groups based on their similarity. Hence, each group is represented by its center that corresponds to the mean of points assigned to the group.

To identify the collaboration profiles of each cluster, we examine the behavior of each pattern according to the four main categories defined in Section 4.3. We can infer 16 different collaboration profiles from such categories, and Table 2 presents their characteristics, each one represented by a network metric. In summary, to define the standard profiles, we use the characteristics to represent the threshold levels: 1 for a high metric value, that is, greater than or equal to 0.5; or 0 for low metric values, that is, less than 0.5. Then, we plot radar charts for each profile, as well as for each detected cluster⁶. Radar charts are extremely suitable for showing outliers and similarities; then, we can clearly identify which profile each cluster belongs to based on the generated plots.

Figure 5 shows the profiles’ characterization of each cluster. Note that *Cluster 1* presents high metric values related to the *Interaction*, *Distance* and *Influence* categories. However, it has an intermediary value for *Similarity*. On the other hand, *Cluster 2* only presents high values for *Distance* and *Similarity*. Finally, *Cluster 3* presents only minimum values for all the categories. Comparing such values with Table 2 allows to match collaboration profiles and clusters.

To summarize, we nominate the identified clusters as follows: *Cluster 1* as Diverse Collaboration (Diverse); *Cluster 2* as Regular Collaboration (Regular); and *Cluster 3* as No Collaboration (Absent). According to the data, the collaborative characteristics of *Diverse* are more similar to profile 1B 2B 3A 4B; the *Regular* cluster is

⁶Visualizations available on the project page at http://bit.ly/comusic_project

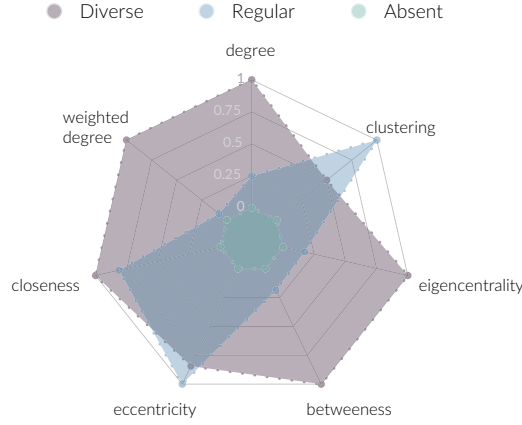


Figure 5: Collaboration profiles of each cluster.



Figure 6: Spearman's Correlation Matrix.

Table 3: Shapiro-Wilk test results. Null hypothesis H_0 : the distributions are Normal

Sample	Statistic	p-value	Normal
Popularity	0.93788	$5.269e^{-11}$	no
Followers	0.6737	$< 2.2e^{-16}$	no
Degree	0.78467	$< 2.2e^{-16}$	no
Weighted Degree	0.76878	$< 2.2e^{-16}$	no
Eccentricity	0.74696	$< 2.2e^{-16}$	no
Closeness	0.78364	$< 2.2e^{-16}$	no
Clustering	0.87115	$< 2.2e^{-16}$	no
Betweenness	0.58553	$< 2.2e^{-16}$	no
Eigencentrality	0.66979	$< 2.2e^{-16}$	no

Table 4: Statistical correlation between: Popularity on Spotify vs. Topological Metrics, and Number of Spotify Followers vs. Topological Metrics, where *** = Strong Correlation; ** = Moderate Correlation; * = Weak Correlation; no star = Very Weak Correlation

Metric	Popularity					
	Pearson		Spearman		Kendall	
	Statistic	P-value	Statistic	P-value	Statistic	P-value
Degree	0.5146218 ***	$< 2.2e^{-16}$	0.5132529 ***	$< 2.2e^{-16}$	0.3813272 *	$< 2.2e^{-16}$
Weighted Degree	0.4774128 **	$< 2.2e^{-16}$	0.4562222 **	$< 2.2e^{-16}$	0.3331673 *	$< 2.2e^{-16}$
Eccentricity	0.3134519 *	$1.647e^{-09}$	0.2151929	$4.457e^{-05}$	0.1623648	$7.095e^{-05}$
Closeness	0.3315823 *	$1.569e^{-10}$	0.4756752 **	$< 2.2e^{-16}$	0.3450073 *	$< 2.2e^{-16}$
Clustering	0.04256444	0.4247	0.1071778	0.04388	0.06908103	0.07314
Betweenness	0.4412301 **	$< 2.2e^{-16}$	0.5091025 ***	$< 2.2e^{-16}$	0.3800098 *	$< 2.2e^{-16}$
Eigencentrality	0.4287696 **	$< 2.2e^{-16}$	0.4875358 **	$< 2.2e^{-16}$	0.3526067 *	$< 2.2e^{-16}$
Metric	Followers					
	Pearson		Spearman		Kendall	
	Statistic	P-value	Statistic	P-value	Statistic	P-value
Degree	0.1997008	0.0001554	0.1849627	0.0004688	0.1307009	0.0004795
Weighted Degree	0.2496251	$1.979e^{-06}$	0.3029554 *	$5.985e^{-09}$	0.2175485	$4.906e^{-09}$
Eccentricity	0.1004326	0.05906	0.02492095	0.6403	0.01792927	0.6521
Closeness	0.1414757	0.007679	0.160929	0.00239	0.1100344	0.002724
Clustering	0.04100568	0.4418	0.07265466	0.1726	0.05100541	0.174
Betweenness	0.1932948	0.0002537	0.2147099	$4.64e^{-05}$	0.1531739	$5.268e^{-05}$
Eigencentrality	0.1589018	0.002716	0.1501669	0.004634	0.1038141	0.004559

more similar to the 1A 2B 3B 4A pattern; and the *Absent* cluster is identical to profile 1A 2A 3A 4A.

5.2 Collaboration Profiles vs. Success

After identifying the collaborations profiles present in the network, we verify if there is a correlation between an artist's success and his/her collaboration behavior. Initially, we present the results of the normality test, because such a method helps to confirm the conclusions on the subsequent analyses. As already mentioned in the Methodology section, we apply the Shapiro-Wilk test [17] with the null hypothesis that the distributions are Normal. According to the results in Table 3, all p-values are much smaller than the significance level of $\alpha = 0.05$, implying that the data distribution is significantly different from a normal distribution. In other words, we cannot assume normality for any of the data samples.

As data is not normally distributed, non-parametric correlations must be applied. Here, we consider Pearson, Spearman and Kendall rank-based correlation tests (see Section 4.4), with statistical correlation evaluation in Table 4. Such tests consider the null hypothesis that there is a correlation between the variables. Considering popularity, the results show only the p-value of the Clustering metric is above $\alpha = 0.05$. Therefore, all three correlation coefficients are statistically significant for all metrics, except for clustering coefficient. On the other hand, considering the number of followers, both the clustering coefficient and the eccentricity reached a high p-value (above $\alpha = 0.05$). That is, we cannot be sure whether there is a correlation between their numbers or not regarding followers.

Figure 6 shows the correlation matrix of the data for Spearman's test. Analyzing only the correlation between popularity and the network metrics, we notice there is a direct association between all the data points. In addition, the tests achieve a moderate and weak (between 0.6 and 0.3) relationship between popularity and most metrics with statistical significance. In fact, only the eccentricity measure exhibited a weaker (less than 0.3) association in all three methods. Regarding the number of followers, all three correlation coefficients are very low. However, we can see a strong correlation between popularity and the number of followers, as expected.

Figure 7 shows an analysis for each cluster related to the success metrics (popularity and number of followers). In the box plot, the

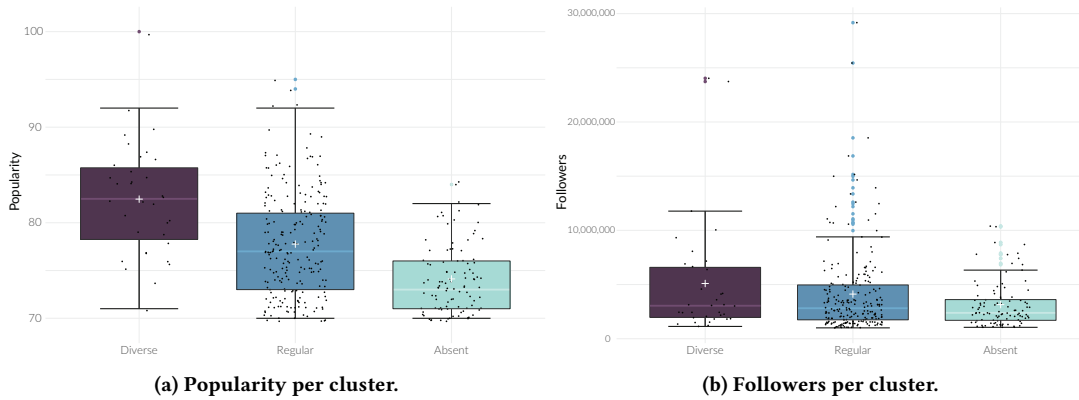


Figure 7: Box plots of each cluster in relation to the success metrics.

central rectangle spans the first quartile to the third quartile; a segment within the rectangle shows the median; and upper and lower stems (whiskers) show the minimum and maximum locations. In summary, the chart identifies where 50% of the most likely values, the median and the extreme values are located. Figures 7a and 7b show the box plots of the popularity's measure and number of followers of each cluster, respectively.

Figure 7a shows that all three clusters have different average levels of popularity, decreasing from *Diverse* to *Absent*. In the first cluster, the mean and median are equivalent to approximately 85. The second one shows the mean and median values less than 80. The last cluster has the lowest rates, close to the minimum values of each cluster (around 70). In fact, 50% of the artists in this cluster are less popular than half of the artists in *Diverse*. We also notice that all three clusters present outliers in their box plots. Nonetheless, the atypical values continue to follow a descending pattern from the first to the third cluster. In addition, in *Absent*, the cluster has less dispersed data, with a uniform popularity distribution.

On the other hand, the clusters in Figure 7b have equivalent median values regarding the number of followers. Nevertheless, with respect to the mean, the rates are also decreasing from *Diverse* to *Absent*. *Diverse* has median values close to 5,000,000 of followers and a maximum value greater than 10,000,000. *Regular* has an average of less than 5,000,000 and a maximum value around to 10,000,000 of followers. *Absent* is the one that presents the smallest mean and maximum value. In fact, 75% of the artists in this cluster have a number of followers less than 2,500,000. As the popularity box plot, the three clusters present outliers. In the opposite way, the outlying values escape the descending pattern previously observed. Particularly, the most atypical data value belongs to the second cluster (29,165,243). In addition, the third cluster also presents the lowest variability and standard deviation, meaning a high predictability.

5.3 Discussion

The results presented in Section 5.1 show the collaboration profiles detected in each cluster of the musical network. We define that highly collaborative profiles are those that have a Degree value greater or equal than 30; profiles with moderate collaboration have between 10 and 30 musical collaborations; and non-collaborative

profiles have less than 10 collaborations. According to the data, the *Diverse* group is characterized by the profile **1B 2B 3A 4B**. In other words, the artists of this cluster have a profile with high interaction (collaborative), a high degree of distant collaborations, a medium level of connections between different genres (diversity), and musical collaborations between important artists (influence). Then, *Regular* presents the profile **1A 2B 3B 4A**. This pattern characterizes artists who are not very collaborative (low interaction), with a high degree of distant collaboration and between the same musical genres (medium diversity), and rare musical collaborations between important artists (low influence). Finally, the artists in *Absent* are characterized by the collaboration profile **1A 2A 3A 4A**; which means this cluster contains non-collaborative artists. In fact, the vast majority of this cluster consists of bands, which usually do not collaborate with other artists.

With the collaboration profiles of each cluster defined, we can discuss their impacts on the artists' success. According to the results in Section 5.2, there is a moderate to strong relationship between the metrics of interaction, proximity and influence, and Spotify's measure of popularity. These results indicate that such metrics may positively impact an artist's popularity. Therefore, successful artists are very likely to present highly collaborative profiles with distant and influential collaborations. We did not find correlation between popularity and the clustering coefficient, thus we cannot infer that the diversity of musical genres impacts on musical success. According to Figure 6, we observed a weak but statistically significant relationship between the interaction and influence measures and the number of followers. However, for the proximity and clustering metrics, no significant relationship was detected. In this case, there is insufficient statistical evidence to support the claim that the number of an artist's followers impacts (positively or negatively) their success. However, in Figure 6, we can see that the number of followers is strongly correlated with the popularity of the artist (which is expected to some degree).

To conclude, with Figure 7 we can validate some of the outcomes achieved. For instance, *Diverse* presents the profile that is more likely of increasing musical success, and this is the exact cluster composed of the most successful artists. In fact, returning to the example cited in our introduction (Section 1), one of the world's

most successful hip-hop artists, Dj Khaled, belongs precisely to the first cluster. In a similar way, *Absent* has the less collaborative profile, which does not necessarily expand musical success. As reported in Figure 7, this is the cluster that presents the lowest values of success measures. Therefore, we conclude that more successful artists are more likely to have a high degree of collaboration between influential and diversified artists. Similarly, those who prefer to pursue a non collaborative musical career may be missing an opportunity to enhance and expand their potential. Moreover, these success characteristics affect more significantly than the musical genre factor. While the first observations may seem intuitively obvious, the latter conclusion is less evident. This is because, typically, highly connected entities tend to achieve some metrics of popularity (regardless of how popularity is defined and what it implies). However, it is counterintuitive to believe that musical genres are not as influential in an artist's success.

Threats to Validity. Despite the relevant results, we cannot forget that correlation does not imply causation. If there is a direct mathematical correlation between variables, it does not indicate a relationship. Other factors can represent the cause and not all success factors were covered here. For instance, the number of awards received by artists, number of views of music clips, the success of the songs, among others. Such external factors can improve the correlation analysis between musical success and collaborative patterns. Another important limitation to be mentioned is the fact that the data source used to collect the artists considers only mainstream and popular music, generalizing the data. Therefore, it would be necessary to include less successful artists in the analysis to conduct good predictive models.

6 CONCLUSION

In this paper, we identified collaboration profiles present in a musical network. Moreover, we analyzed the relationship between such collaborative patterns and the artists' success. Specifically, using data from Billboard and Spotify, we modeled a social network and defined a measure of success. Then, to identify the collaborative profiles, we applied six topological metrics and defined four main categories: Interaction, Distance, Similarity and Influence. Then, using the K-Means algorithm, we identified three clusters with distinct collaboration profiles. To measure the relationship between these profiles and musical success, we conducted a statistical analysis.

Through the outcomes of an extensive data analysis, we identified moderate and statistically significant correlations between popularity and measures of degree and centrality. However, we found no substantial relationship in terms of the number of followers and social metrics. Our results provide strong evidence that clusters with a high degree of interaction, influence, and diversity, are more likely to present successful artists. Moreover, these success characteristics affect more significantly than the musical genre factor. Another valuable contribution of this study is that our exploratory approach based on collaborative networks can easily be extended to other areas of knowledge (e.g., arts and science).

As future work, we plan to conduct a more accurate analysis on a shorter scale by exploring other metrics for artistic success. Moreover, we are studying other possibilities to help establish causality relationships.

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